

	(c)	line from origin, curved in correct direction but not level out	B1	[1]
6	(a)	(i) 26 protons	B1	
		(ii) 30 neutrons	B1	[2]
	(b)	(i) mass = $56 \times 1.66 \times 10^{-27}$	C1	
		(allow $\times 1.67 \times 10^{-27}$ but 0/2 for use of 26 or 30)		
		= 9.3×10^{-26} kg	A1	
		(ii) density = mass/volume where volume = $\frac{4}{3} \times \pi \times r^3$	C1	
		= $(9.3 \times 10^{-26}) / (\frac{4}{3} \times \pi \times \{5.7 \times 10^{-15}\}^3)$		
		= 1.2×10^{17} kg m ⁻³	A1	[4]
	(c)	nucleus occupies only very small fraction of <u>volume of atom</u>		
		or 'lot of empty space inside atom'	B1	
		(do not allow spacing between atoms)		
		any further good physics e.g. nuclear material is very dense	B1	[2]
7	(a)	(i) $P = Vi$	C1	